



“To Study Awareness & Use of Headphones of Bus Passengers In Satara City.”

A FIELD PROJECT SUBMITTED TO,

KARMAVEER BHAURAO PATIL UNIVERSITY, SATARA

FOR THE COURSE OF

B.sc III IN

STATISTICS

BY

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2025-26

DECLARATION

We hereby declare that the Field Project entitled, “To Study Awareness & Use of Headphones of Bus Passengers In Satara City” completed and written by our group, has not formed the basis for the award of any degree or similar title of this or any other university or examination body. Further, I declare that I have not violated any of the provisions under the acts of copyright/Piracy/Cyber/IPR, etc, amended from time to time.

The present Field work is original, and the conclusions drawn therein are based on the analysis done by ourselves.

Place: Satara

Date:



Rayat Shikshan Sanstha's
Yashwantrao Chavan Institute of Science, Satara
(Constituent College of KBP University, Satara)

CERTIFICATE

Department of Statistics

Date:

This is to certify that the following students are working under my guidance. & partially fulfilled of curriculum of B.sc. III students has successfully completed the project work in statistics entitled "To Study Awareness & Use of Headphones of Bus Passengers In Satara City" as prescribed by the Karmaveer Bhaurao Patil University, Satara, during the academic year 2025-26.

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ACKNOWLEDGEMENT

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To Study Awareness & Use of Headphones of Bus Passengers In Satara City.

❖ Introduction of project :

In today's daily life the use of headphones has become very common especially during travel. Many bus passengers use headphones to listen to music, attend calls, or pass time while traveling. Headphones provide comfort and entertainment but their improper and excessive use can cause health problems such as ear pain, headache, and hearing loss.

In a city like Satara, a large number of people travel by bus every day for work, education, and other purposes. During travel many passengers use headphones for long periods without knowing the safe volume level and correct duration of use. Most users are not fully aware of the harmful effects of continuous headphone use and the importance of safe listening habits.

This study, "To Study Awareness and Use of Headphones of Bus Passengers in Satara City," aims to understand how often passengers use headphones during travel and how much awareness they have about its effects on health. The study focuses on usage pattern and awareness level among bus passengers.

The findings of this study will help to identify the level of awareness among passengers and highlight the need for promoting safe and responsible use of headphones during travel.

❖ Harmful Effects of Headphone Use on Health

- **Hearing Loss:** Listening to headphones at high volume for a long time can reduce hearing ability.
- **Ear Pain:** Continuous use of headphones can cause pain and discomfort in the ears.
- **ringing in Ears:** Excessive headphone use may cause a ringing or buzzing sound in the ears.
- **Headache:** Loud sound and long duration of use can lead to frequent headaches.
- **Low Awareness During Travel:** Headphones reduce awareness of surrounding sounds, increasing the risk of accidents.
- **Ear Infection:** Using dirty or shared headphones can cause ear infections.
- **Mental Stress:** Continuous sound exposure can cause mental stress and tiredness.
- **Sleep Problems:** Overuse of headphones can disturb sleep patterns.
- **Neck Pain:** Long hours of headphone use in a wrong posture can cause neck pain.
- **Addiction:** Regular and long-term use of headphones can lead to dependency

OBJECTIVES

❖ Objectives of the Study :

1. **To study the use of headphones among passengers in Satara City:-**This objective focuses on understanding how commonly headphones are used by bus passengers during travel in Satara City.
2. **To analyze the duration and pattern of headphone use during travel :-**This objective aims to study how long and in what manner passengers use headphones while traveling.
3. **To examine awareness about mobile headphone warnings:-**This objective focuses on finding out whether passengers are aware of mobile and headphone-related warning messages.
4. **To understand health problems related to headphone use:-**This objective aims to identify common health issues faced by passengers due to headphone use.
5. **To study the impact of headphone use on travel safety:-** This objective focuses on understanding how headphone use affects alertness and safety during travel.

COLLECTION OF DATA

The data for the present study was collected from **Satara Bus Stand**. The bus stand was selected as the study area because a large number of passengers travel daily making it suitable for collecting reliable information related to headphone use during travel.

The data was collected from bus passengers who were waiting for buses or traveling at the Satara Bus Stand. Primary data was collected directly from passengers using a **structured questionnaire**. The questionnaire included simple and clear questions related to the use of headphones, duration of use, awareness about mobile headphone warnings, health problems, and travel safety.

The researcher personally visited the Satara Bus Stand and explained the purpose of the study to the respondents before collecting the data. Only willing passengers were included in the study. Care was taken to ensure that the responses were honest and accurate.

The data was collected during different time periods to get a better and balanced response from passengers. The collected data was then properly arranged and used for further analysis and interpretation.

Statistical Summary

For the present study, a combination of graphical methods and statistical tests was used to analyze the data collected from bus passengers at Satara Bus Stand. These tools helped in clear presentation, comparison, and interpretation of data.

1. Diagrammatic Representation of Data

To present the data in a simple and understandable form, the following diagrams were used:

Pie Chart: Used to show the proportion of headphone users and non-users and other categorical responses.

Bar Diagram: Used to compare different groups such as age groups and awareness levels.

2. Statistical Tests Used

1. Descriptive Analysis (Frequency and Percentage): This method is used to summarize the data in a simple and meaningful way.

2. Chi-Square Test of Association: This test is used to examine whether there is a significant association between two categorical variables.

3. Single Proportion Test: It is a statistical method used to determine whether the proportion observed in a sample is significantly different from a specified population proportion.

4. Mann-Whitney U Test: This non-parametric test is used to compare two independent groups when the data is ordinal

5. Kruskal-Wallis Test: This test is used to compare more than two independent groups when the data is ordinal.

Methodology

❖ Methodology :

The present study is based on a descriptive research design. The main objective of the study is to examine the awareness and usage of headphones among bus passengers in Satara City.

1. Study Area

The study was conducted in Satara City, focusing on bus passengers traveling by local and long-distance buses.

2. Sample Size

A total of 380 bus passengers were selected for the study.

3. Sampling Method

The respondents were selected using the convenience sampling method, as data was collected from passengers who were easily available at bus stands and inside buses.

4. Data Collection

The study is mainly based on primary data. Data was collected through a structured questionnaire prepared according to the objectives of the study.

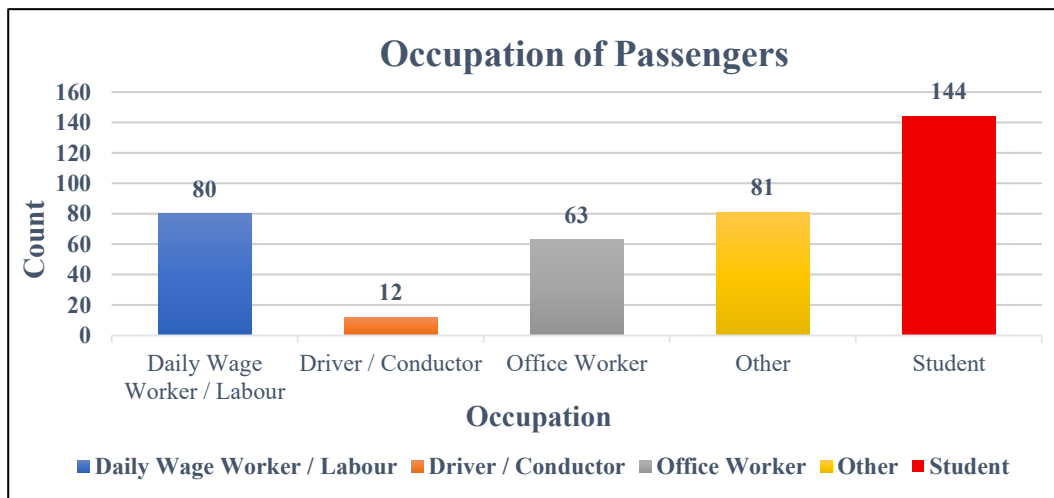
Secondary data was collected from research articles, journals and online sources related to headphone use and travel safety

5. Data Presentation

The analyzed data was presented in the form of tables, charts and graphs for better understanding and interpretation.

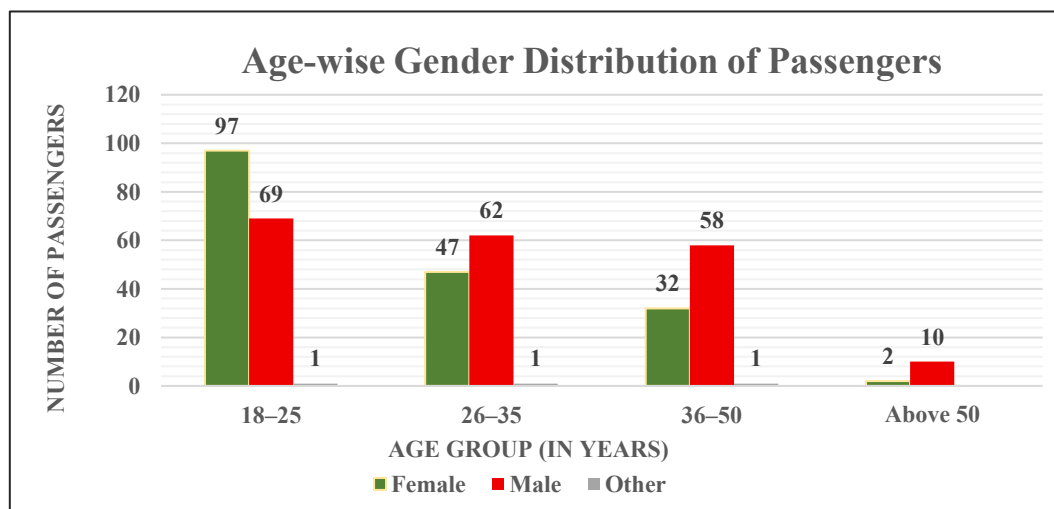
DATA VISUALISATION

❖ Demographic Profile of Headphone Users :



Interpretation :

Students form the largest group (37.9%), followed by other occupations (21.3%) and daily wage workers (21.1%). Drivers/conductors account for the smallest share (3.2%).

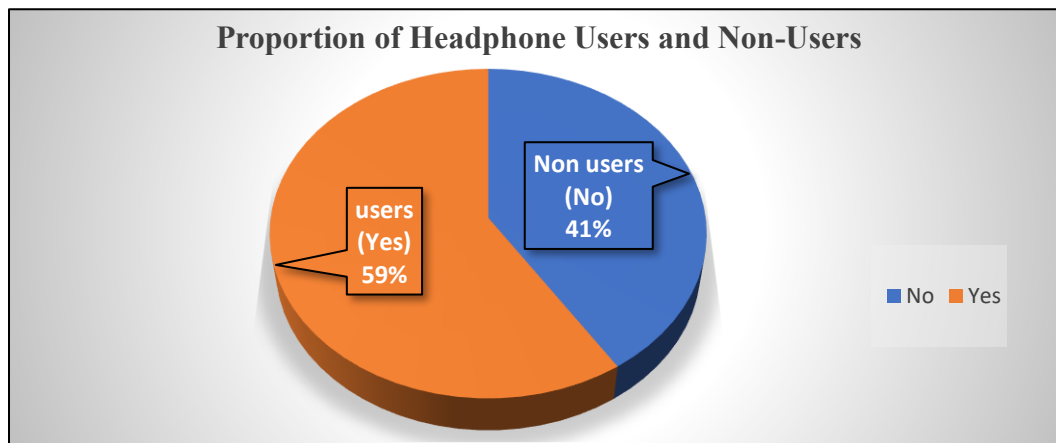


Interpretation :

Males constitute 52.4% and females 46.8% of passengers. The 18–25 age group (43.9%) has the highest share, while passengers above 50 years (3.2%) are least represented.

Conclusion : Students and young passengers (18–25 years) form the majority of headphone users. Male and female participation is almost balanced, while others age groups and drivers/conductors show the least response.

❖ Proportion of Headphone Users and Non-Users

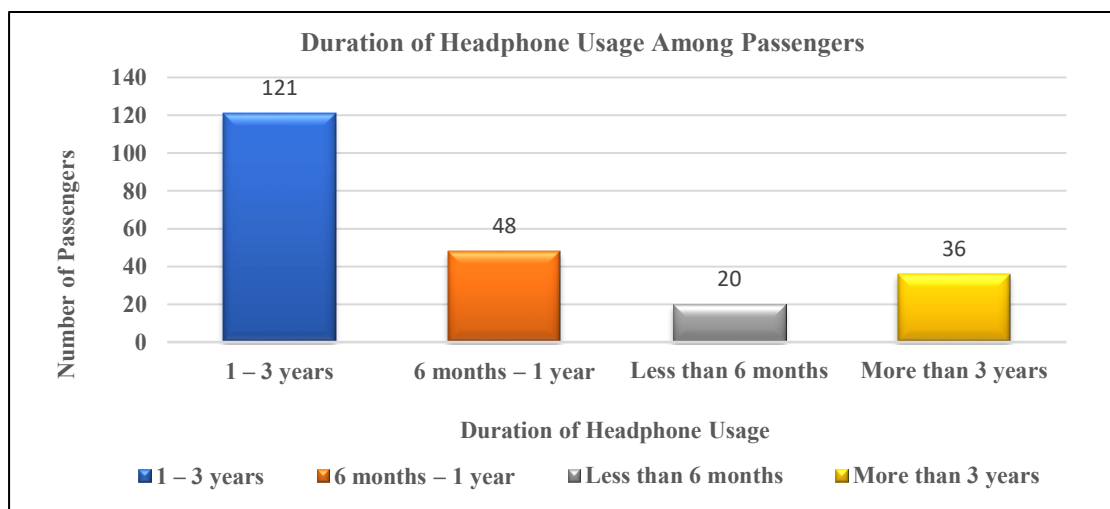


Headphone users / non users	Count of passengers in city
No (Non Users)	155
Yes (Users)	225
Grand Total	380

Interpretation :

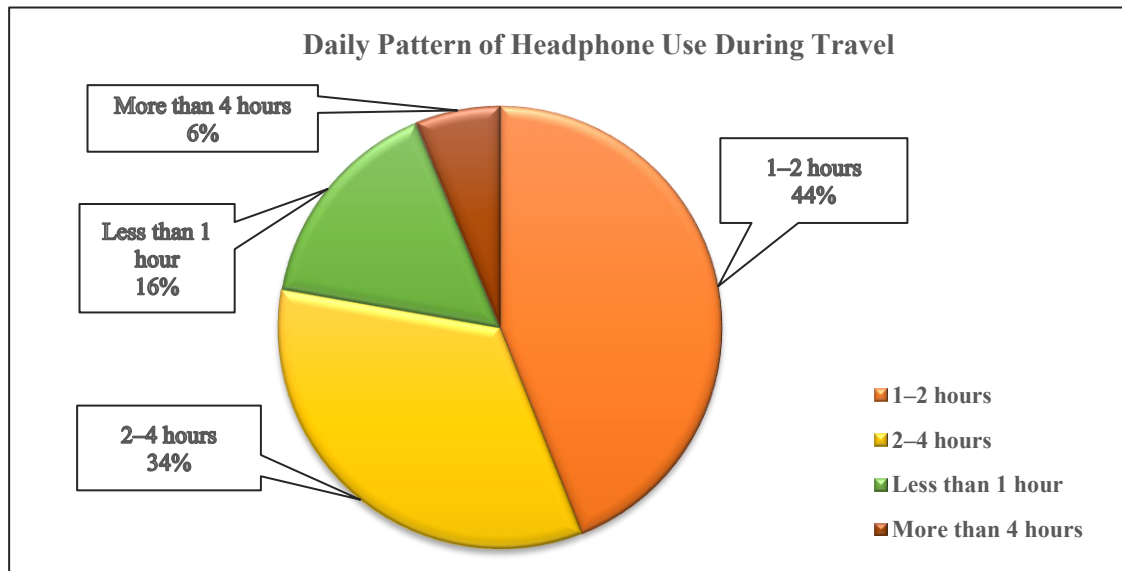
Out of 380 passengers surveyed in Satara City, **225 passengers (59%)** reported using headphones, while **155 passengers (41%)** reported not using headphones. The pie chart clearly shows that headphone users form a larger proportion of the sample compare to headphone non users .

❖ To analyze the duration and pattern of headphone use during travel



Interpretation:

Out of 225 headphone users, 121 (53.8%) have been using headphones for 1–3 years, representing the largest group. This is followed by 48 (21.3%) users for 6 months–1 year, 36 (16.0%) users for more than 3 years, and 20 (8.9%) users for less than 6 months. The data clearly indicates that most passengers are experienced headphone users.



Average time per Day	Count of Daily headphone users
1-2 hours	99
2-4 hours	76
Less than 1 hour	36
More than 4 hours	14
Grand Total	225

Interpretation:

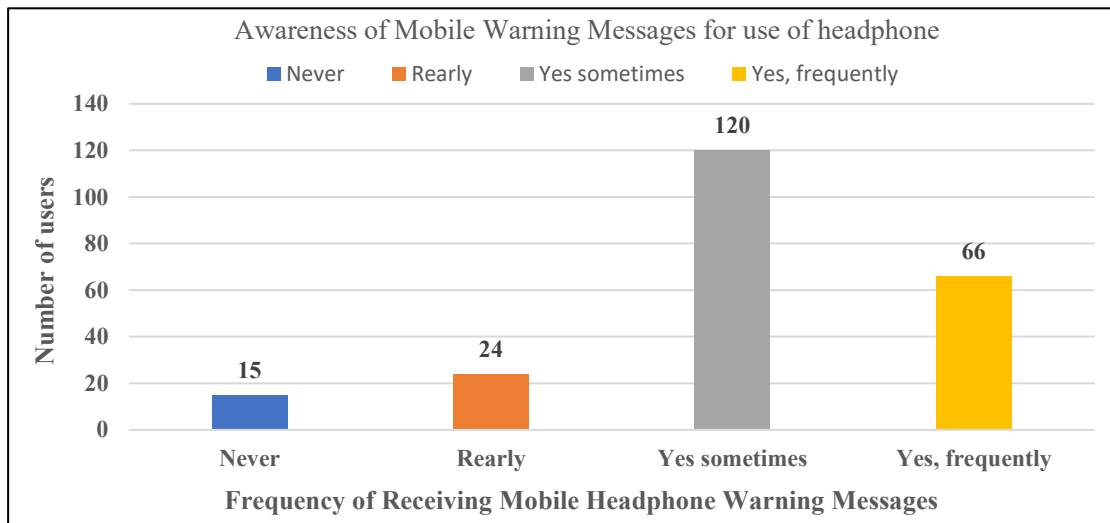
Out of 225 headphone users, **99 passengers (44.0%)** use headphones for **1-2 hours per day**, which forms the largest share. This is followed by **76 passengers (33.8%)** using headphones for **2-4 hours**, **36 passengers (16.0%)** using them for **less than 1 hour**, and **14 passengers (6.2%)** using headphones for **more than 4 hours per day**.

The pie chart indicates that most passengers use headphones for a moderate duration on a daily basis.

Conclusion

The study shows that most passengers have been using headphones for a long period and use them for a moderate number of hours each day. This indicates that headphone use is a regular habit among passengers, but daily usage is generally not for very long hours.

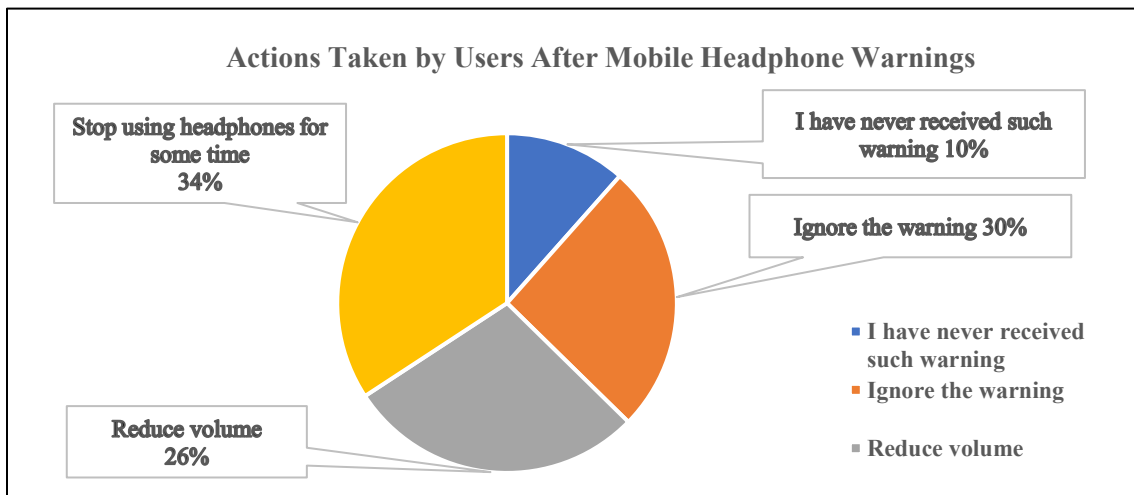
❖ Awareness of Mobile Headphone Warning Messages :



Interpretation :

Out of 225 passengers, 120 (53.3%) reported receiving warning messages sometimes, while 66 (29.3%) receive them frequently. In contrast, 24 passengers (10.7%) reported receiving warnings rarely, and only 15 passengers (6.7%) reported never receiving such warnings. This shows that the majority of passengers are aware of mobile warnings related to high volume or excessive headphone use.

❖ Actions Taken by Users After Mobile Headphone Warnings:



Actions Taken by Users	Count of warning receivers
I have never received such warning	15
Ignore the warning	69
Reduce volume	64
Stop using headphones for some time	77
Grand Total	225

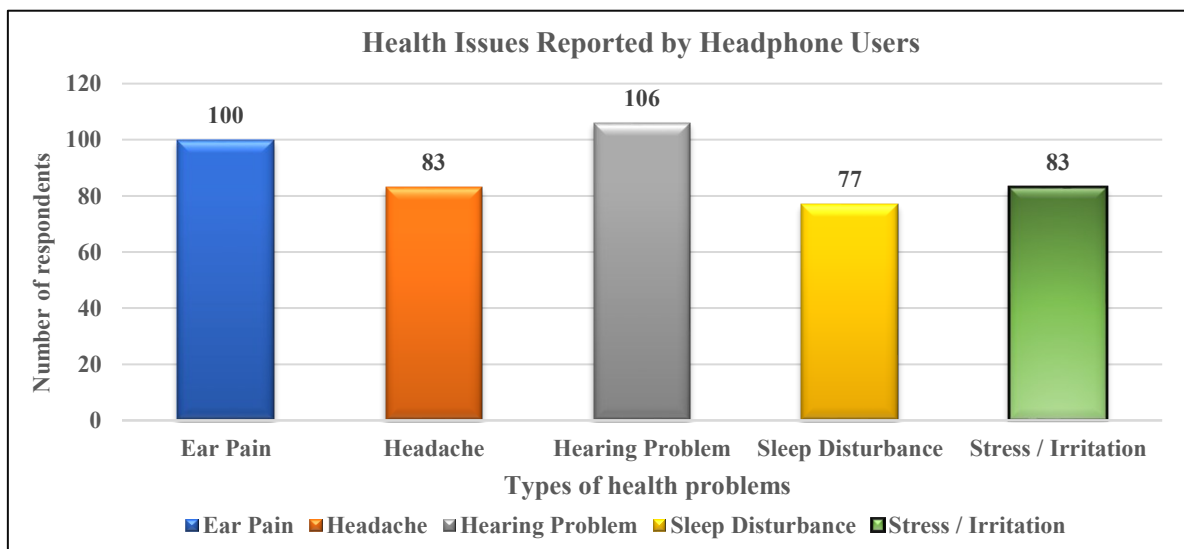
Interpretation:

Out of 225 users, 77 users stopped using headphones for some time after receiving a warning, which is the most common action. This is followed by 69 users who ignored the warning and 64 users who reduced the volume, while 15 users reported that they had never received such a warning. The data indicates varied user responses to mobile headphone warnings.

Conclusion :

The study concludes that mobile headphone warnings influence user behaviour to some extent. While many users take corrective actions such as stopping headphone use temporarily or reducing volume, a noticeable number of users still ignore the warnings.

❖ Health Problems Experienced Due to Headphone Use:

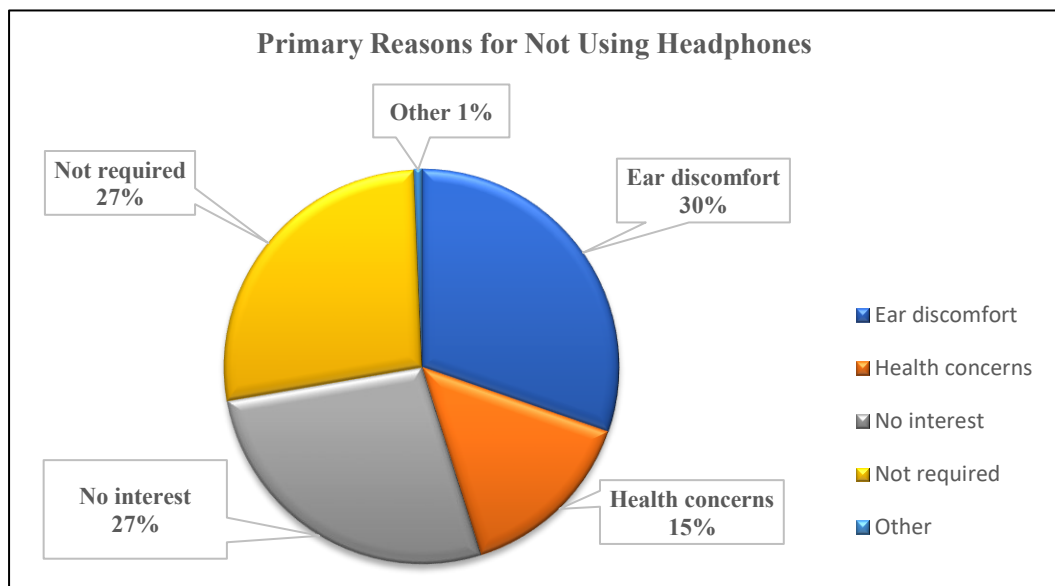


Health problem	Count of users
Ear Pain	100
Headache	83
Hearing Problem	106
Sleep Disturbance	77
Stress / Irritation	83
Grand Total	449

Interpretation :

"The graph represents various health issues reported by headphone users. Among 225 users, Hearing Problems were the most common (106; 47.1%), followed by Ear Pain (100; 44.4%). Headache and Stress/Irritation were reported by 83 users (36.9%) each, while Sleep Disturbance was reported by 77 users (34.2%). Since multiple responses were allowed, the total number of responses (449) exceeds the actual number of users."

❖ Distribution of Non-Users by Reason for Avoidance:



Reason for not using headphone	Numbers of passengers
Ear discomfort	47
Health concerns	23
No interest	42
Not required	42
Other	1
Grand Total	155

Interpretation:

"The pie chart illustrates the reasons why 155 passengers do not use headphones. A majority of the non-users (**58.1%**) stated that they simply do not feel the necessity or interest in using them. However, a significant **29%** avoid headphones due to health concerns or medical advice, while the remaining **12.9%** stopped using them due to past discomfort or bad experiences. This indicates that while some avoid headphones by choice, others do so to prevent or manage auditory health issues.

Conclusion (English):

"The analysis of non-users reveals that a substantial portion of the population (nearly **42%**) avoids headphones due to health-related reasons or past discomfort. This highlights a conscious effort among some passengers to prioritize their ear health over the convenience of using personal audio devices during travel."

STATISTICAL DATA ANALYSIS

Cochran's Sample Size Formula

For an unknown or infinitely large population, uses the standard **Cochran's formula** for infinite populations.

For a finite and known population, it applies the [finite population correction \(FPC\)](#) to compute the adjusted sample size.

In addition to providing the minimum sample size, the calculator also gives a step-by-step solution of how the sample size was calculated, making it easy to understand and apply for our research.

Sample Size Determination

The present study was conducted at Satara Bus Stand over a period of **seven** days. Based on information obtained from the depot authorities, it was estimated that approximately 3000 passengers travel daily through the bus stand. Therefore, the total population of passengers during the study period was assumed to be around **21,000**.

Confidence level (%)

Example: 95 for 95% confidence level.

Margin of error (e)

Enter as decimal (0.05 = 5%).

Estimated proportion (p)

Enter as decimal (0.5 = 50%). Defaults to 0.5 (maximum variability).

Population size (N) (optional)

Recommended sample size: 378

Step-by-step Solution

$$n_0 = \frac{Z^2 p(1 - p)}{e^2}$$

$$n_0 = \frac{(1.9600)^2 \times 0.5000 \times 0.5000}{(0.0500)^2}$$

$$= 384.145882$$

Finite population correction:

$$n = \frac{384.145882}{1 + \frac{384.145882 - 1}{21000}}$$

$$= 377.262708$$

$$n \approx 378$$

Conclusion :

Thus, the required sample size for the study was calculated to be approximately **378 passengers**. For practical convenience and better representation, the sample size was rounded to **380 passengers**. These respondents were selected from bus passengers over the seven-day survey period.

❖ Descriptive Analysis (Frequency and Percentage):

1. Introduction

This statistical survey presents a detailed statistical analysis of the data collected for the study on **headphone usage behaviour and awareness**.

2. Final Sample Size:

N = 380 respondents

3. Classification of passengers in (percentage %)

Based on the passengers response to headphone usage, the passengers were classified into two groups.

Headphone users / non users	Frequency	Percentage %
No (Non users)	155	40.8%
Yes (Users)	225	59.2%
Grand Total	380	100%

$$Percentage = \frac{Frequency}{Total} * 100$$

Total passengers = 380

- **Users**

$$Percentage = \frac{225}{380} * 100 = 59.2\%$$

- **Non-Users**

$$Percentage = \frac{155}{380} * 100 = 40.8\%$$

Conclusion :

Out of 380 passengers, 59.2% use headphones, while 40.8% do not use headphones.

4. Age-wise Gender Distribution of Passengers :

Age of passengers	Gender			
	Female	Male	Other	Grand Total
18–25	97	69	1	167
26–35	47	62	1	110
36–50	32	58	1	91
Above 50	2	10		12
Grand Total	178	199	3	380

5. Age-wise Distribution of Headphone Usage

To study the relationship between age group and headphone users and non users passengers were divided into:

- 18–25 years
- 26–35 years
- 36–50 years
- Above 50 years

Age Group vs Headphone Users and non users (Observed Frequencies)

Age of passengers	Count of headphone users /non users		
	No (non users)	Yes (users)	Grand Total
18–25	32	135	167
26–35	48	62	110
36–50	65	26	91
Above 50	10	2	12
Grand Total	155	225	380

❖ Risk Analysis of Headphone Usage

Concept of Risk :

In this study, risk refers to the proportion (probability) of passengers using headphones within a specific age group.

$$\text{Risk} = \frac{\text{Number of headphone users in the group}}{\text{Total respondents in the group}}$$

Risk of Headphone Users (Yes):

Risk Calculation

- 18–25

$$\text{Risk} = 135 / 167 = 0.808 = 80.8\% \text{ (chance)}$$

- 26–35

$$\text{Risk} = 62 / 110 = 0.563 = 56.3\% \text{ (chance)}$$

- 36–50

$$\text{Risk} = 26 / 91 = 0.285 = 28.5\% \text{ (chance)}$$

- Above 50

$$\text{Risk} = 2 / 12 = 0.167 = 16.7\% \text{ (chance)}$$

Conclusion:

The study shows that the **chance (probability)** of headphone usage is highest among the **18–25 age group (80.8%)** and gradually decreases with increasing age. This indicates that younger individuals are more likely to use headphones compared to older age groups.

Probability of Headphone non users (No):

$$\text{Probability} = \frac{\text{Number of headphone non users in the group}}{\text{Total respondents in the group}}$$

Probability Calculation:

- 18–25

$$32 / 167 = 0.192 = 19.2\% \text{ (chance)}$$

- 26–35

$$48 / 110 = 0.436 = 43.6\% \text{ (chance)}$$

- 36–50

$$65 / 91 = 0.714 = 71.4\% \text{ (chance)}$$

- Above 50

$$10 / 12 = 0.833 = 83.3\% \text{ (chance)}$$

Interpretation:

The chance of non-users increases with age. The highest non-usage is observed in the Above 50 age group (83.3%), while the lowest is in the 18–25 age group (19.2%).

Final Conclusion:

The analysis shows that headphone usage is highest among the 18–25 age group (80.8%), while non-usage is highest among the Above 50 age group (83.3%). This clearly indicates that younger individuals are more likely to use headphones, whereas older individuals are less likely to use them.

❖ Frequency Tables :

Frequency Distribution of passengers According to the Main Purpose of Using Headphones (Multiple Responses)

Purpose of use Headphone	Frequency	Percentage
Gaming:	19	8.44 %
Phone calls:	43	19.11 %
watching video	72	32.00 %
online study	83	36.88 %
While traveling:	96	42.66 %
music	99	44.00 %

Note: Total percentage exceeds 100% because multiple responses were allowed.

Total Respondents (N) = 225

Result:

The analysis shows that the majority of passengers use headphones primarily for listening to music (44%), followed by while traveling (42.67%). A considerable number of passengers also use headphones for online work or study (36.89%) and watching videos (32%). A smaller proportion use headphones for phone calls (19.11%), while gaming (8.44%) is the least common purpose among passengers.

Conclusion:

Listening to music is the most common purpose of headphone usage among passengers, followed by travel-related use and online study activities. Gaming is comparatively the least preferred purpose. The findings indicate that entertainment and travel are the dominant reasons for headphone usage.

❖ SINGLE PROPORTION TEST Z-TEST (FOR NON-USERS):

Objective: To examine whether more than 50% of passengers have used headphones in the past.

Category	Have you ever used headphones in the past?
No	60
Yes	95
Grand Total	155

Sample Proportion:

Total sample (n)=155

Headphone use in past (x)=95

$$p = \frac{95}{155} = 0.613$$

So, 61.3% of passengers reported that they have used headphones in the past.

Although the sample proportion (61.3%) is greater than 50%, we cannot directly conclude that the population proportion is greater than 50%.

This is because the sample result may vary due to random sampling error.

Hypothesis testing helps us to determine whether the observed difference (61.3% – 50%) is:

- Due to random chance or
- Statistically significant and reflects the true population proportion.

Hypothesis

Null Hypothesis (H₀):

The proportion of passengers who have used headphones in the past is equal to 50%.

$$H_0: p = 0.5$$

Alternative Hypothesis (H₁):

The proportion of passengers who have used headphones in the past is greater than 50%.

H1: $p > 0.5$

This is a right-tailed test.

Test Statistic Formula :

$$Z = \frac{\bar{p} - p}{\frac{\sqrt{p_0(1-p_0)}}{n}}$$

Where:

$\hat{p}$ = Sample proportion

p_0 = Hypothesized proportion (0.5)

n = Sample size

Calculation :

$$Z = \frac{0.613 - 0.5}{\frac{\sqrt{0.5 \times 0.5}}{\sqrt{155}}} = \frac{0.113}{\sqrt{\frac{0.25}{155}}} = \frac{0.113}{\sqrt{0.00163}}$$
$$= \frac{0.113}{0.040}$$

$$Z = 2.82$$

Decision Rule

At 5% level of significance (one-tailed test):

Critical Z value = 1.645

Since,

$$2.82 > 1.645$$

We reject the null hypothesis H_0 .

Result

The calculated Z value (2.82) is greater than the critical value (1.645). Therefore the result is statistically significant.

Conclusion

There is sufficient statistical evidence at the 5% level of significance to conclude that more than 50% of passengers have used headphones in the past.

❖ **Chi-square (χ^2) test of independence:**

Types of headphones used among passengers by occupation (n = 225)

Statistical test used

To examine the association between occupation and type of headphones used, the Chi-square (χ^2) test of independence was applied.

Distribution of headphone types used by occupation(O_i)

Occupation	Type of Headphone			
	Bluetooth earbuds	Wired	Wireless	Grand Total
Daily Wage Worker / Labour	19	13	9	41
Driver / Conductor	1	2	3	6
Office Worker	17	10	18	45
Other	10	4	5	19
Student	66	23	25	114
Grand Total	113	52	60	225

Hypothesis:

H₀ (Null Hypothesis):

Occupation and type of headphones used are independent. (No association)

H₁ (Alternative Hypothesis):

Occupation and type of headphones used are dependent. (There is association)

Formula of Chi-Square Test Statistic

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Where,

- O_i = Observed frequency
- E_i = Expected frequency

Formula for Expected Frequency (E_i):

$$E_i = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$$

Expected (Ei) values:

Occupation	Type of Headphone		
	Bluetooth earbuds	Wired	Wireless
Daily Wage Worker / Labour	20.58	9.48	10.93
Driver / Conductor	3.01	1.39	1.60
Office Worker	22.60	10.40	12.00
Other	9.54	4.39	5.07
Student	57.24	26.35	30.40

O_i – E_i)² / E_i Calculation Table:

Occupation	Type of Headphone		
	Bluetooth earbuds	Wired	Wireless
Daily Wage Worker / Labour	0.12	1.31	0.35
Driver / Conductor	1.34	0.27	1.23
Office Worker	1.39	0.2	3.00
Other	0.02	0.03	0.0008
Student	1.34	0.43	0.96

Chi-Square Test Statistic:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 11.81$$

$$\chi^2 = 11.81$$

Degree of Freedom (df)

$$df = (r-1)(c-1)$$

Where,

r = 5 rows

c = 3 columns

$$df = (5-1)(3-1)$$

$$df = 4 \times 2 = 8$$

Chi-Square Table Value:

At 5% level of significance ($\alpha = 0.05$)

df = 8

$\chi^2_{table} = 15.507$

Decision Criteria

If χ^2 calculated $> \chi^2$ table $\rightarrow$ Reject H_0

If χ^2 calculated $< \chi^2$ table $\rightarrow$ Accept H_0

Here,

$11.81 < 15.507$

So, **Accept H_0**

Result

There is **no significant association** between occupation and type of headphones used.

Conclusion

From the chi-square test, we conclude that occupation and type of headphones used are independent. This means headphone type preference does not significantly differ according to occupation.

❖ Kruskal–Wallis Test:

To examine the difference in high volume listening levels across age groups.

Introduction

The Kruskal–Wallis test is a non-parametric statistical method used to compare the median values of an ordinal variable across three or more independent groups. It is an alternative to one-way ANOVA when the assumptions of normality are not satisfied.

In this project, the Kruskal–Wallis test is applied to determine whether high-volume listening behavior differs significantly among different age groups. Since the volume level variable is measured on an ordinal scale (Never, Rarely, Sometimes, Always), this test is appropriate.

Variables

Independent Variable: Age Group

- 18–25
- 26–35
- 36–50
- Above 50

Dependent Variable: High Volume Listening Level (Ordinal)

1 = Never

2 = Rarely

3 = Sometimes

4 = Always

Total Sample Size (N) = 225

Formula of Kruskal–Wallis Test

$$H = \frac{12}{N(N+1)} \sum \frac{R^2}{n} - 3(N+1)$$

Where:

H = Kruskal–Wallis test statistic

N = Total number of observations

R = Sum of ranks for each group

n = Sample size of each group

The calculated H value follows a chi-square distribution with (k – 1) degrees of freedom.

Hypothesis

H₀: There is no significant difference in median high-volume listening levels among age groups.

H₁: At least one age group differs significantly in median high-volume listening level.

R Code:

```
age <- c(rep("18-25", 135),
        rep("26-35", 62),
        rep("36-50", 26),
        rep("Above50", 2))

volume <- c(rep(4, 20), rep(1, 7), rep(2, 17), rep(3, 91),
            rep(4, 23), rep(1, 1), rep(2, 8), rep(3, 30),
            rep(4, 9), rep(1, 1), rep(2, 11), rep(3, 5),
            rep(1, 2))
data <- data.frame(age, volume)

obs_table <- table(data$age, data$volume)

data$rank <- rank(data$volume)

rank_summary <- aggregate(rank ~ age,
                           data,
                           function(x) c(N = length(x),
                                           Mean_Rank = mean(x)))

rank_summary <- do.call(data.frame, rank_summary)

kw <- kruskal.test(volume ~ age, data = data)

H_stat <- round(kw$statistic, 2)
df <- kw$parameter
p_value <- kw$p.value

summary_table <- data.frame(Test_Statistic_H = H_stat,
                             Degrees_of_Freedom = df,
                             P_Value = p_value)

obs_table
rank_summary
summary_table
```

Output :

Observed Frequency Table:

Observed Frequency Table:

	1	2	3	4
18-25	7	17	91	20
26-35	1	8	30	23
36-50	1	11	5	9
Above50	2	0	0	0

(1=Never, 2=Rarely, 3=Sometimes, 4=Always)

Rank Summary Table:

Rank Summary Table:

	age	rank.N	rank.Mean_Rank
1	18-25	135	108.0667
2	26-35	62	131.3790
3	36-50	26	103.0192
4	Above50	2	6.0000

Kruskal-Wallis Test Output:

Kruskal-Wallis chi-squared = 54.82

df = 3

p-value < 2.2e-16

Final Test Summary Table:

Hypothesis:

H0: No significant difference in median volume levels among age groups

Alternative:

H1: At least one age group differs in median volume level

Test_Statistic_H: 54.82

Degrees_of_Freedom: 3

P_Value: < 0.001

Decision: Reject H0

Conclusion:

There is a significant difference in volume levels across age groups.

Interpretation:

The analysis indicates a statistically significant difference in high-volume listening behavior across age groups. Younger respondents (18–25 and 26–35) demonstrate higher mean ranks, suggesting a greater tendency to listen at higher volumes. In contrast, older age groups (36–50 and Above 50) show lower median listening levels.

Conclusion

The Kruskal–Wallis test confirms that age group significantly influences high-volume listening behavior. Younger individuals are more likely to engage in high-volume listening compared to older individuals. Therefore, age plays a significant role in determining risky listening behavior patterns.

❖ Overall Conclusion of Data Analysis

Based on the statistical analysis of the data collected from 380 bus passengers in Satara city, the following conclusions are drawn:

- **Prevalence of Usage:** A significant majority of passengers (59.2%) are active headphone users. Among the non-users, the Z-test confirmed that more than 50% had used headphones in the past, indicating a high lifetime prevalence.
- **Demographic Trends:** The Kruskal-Wallis test and descriptive data suggest that the younger generation (age 18-25), primarily students, are the most frequent users of headphones, often using them for more than 2 hours daily.
- **Purpose of Use:** The primary motives for using headphones are listening to music and watching videos during travel, followed by attending online studies or work.
- **Health Impact:** There is a clear indication of physical discomfort among users. Approximately 27.9% of the total respondents reported hearing problems, while a significant number experienced ear pain and headaches. This suggests a lack of awareness regarding safe volume levels.
- **Volume Awareness:** Although most modern smartphones provide high-volume warnings, a substantial portion of users either ignore these warnings or only temporarily reduce the volume, which contributes to long-term auditory risks.
- **Statistical Independence:** The Chi-Square test revealed that the type of headphones used (Wired vs. Wireless/Earbuds) is independent of the user's

occupation, suggesting that the choice of device is more about personal preference than professional necessity.

❖ Recommendations

To mitigate the health risks associated with prolonged headphone use, the following measures are recommended:

- **Adherence to the 60/60 Rule:** Users should be encouraged to follow the 60/60 rule—listening at no more than 60% of the maximum volume for no longer than 60 minutes at a time.
- **Use of Noise-Cancelling Headphones:** Since bus passengers often increase volume to drown out traffic noise, using Active Noise Cancellation (ANC) headphones can help them listen clearly at lower, safer volumes.
- **Public Awareness Campaigns:** Local authorities or health organizations should conduct awareness programs in educational institutes and public transport hubs regarding "Noise-Induced Hearing Loss" (NIHL).
- **Ear Hygiene and Breaks:** Users should take frequent "ear breaks" (at least 10-15 minutes every hour) to allow the ears to recover from sound pressure.
- **Monitoring Safety Features:** Smartphone users should be educated to never disable volume limiters and to take "High Volume Warnings" seriously rather than dismissing them.

❖ References

1. **World Health Organization** (2015). *Make Listening Safe: A Public Health Approach to Hearing Loss*. Geneva: WHO Press. This report explains the risks of unsafe listening practices, safe volume levels, and preventive measures for hearing protection.
2. **World Health Organization** (2021). *World Report on Hearing*. Geneva: WHO. This global report discusses the increasing burden of hearing problems due to personal audio device usage and emphasizes awareness and early prevention.
3. **Centers for Disease Control and Prevention** (2022). *Noise and Hearing Loss Prevention*. Atlanta, GA: CDC. This report highlights the health effects of prolonged exposure to loud sounds and provides guidelines for safe listening habits.
4. **National Institute on Deafness and Other Communication Disorders** (2023). *Noise-Induced Hearing Loss*. This publication explains how continuous exposure to high sound levels through headphones can damage hearing permanently.

QUESTIONNAIRE:

Title: Survey on Awareness & Use of Headphones among Bus Passengers in Satara City

Section A: Demographic Information (For All Passengers)

1. Age Group

- ☐ 18–25
- ☐ 26–35
- ☐ 36–50
- ☐ Above 50

2. Gender

- ☐ Male
- ☐ Female
- ☐ Other

3. Occupation

- ☐ Student
- ☐ Office Worker
- ☐ Daily Wage Worker / Labour
- ☐ Driver / Conductor
- ☐ Other

Section B: Headphone Usage

4. Do you use Headphones / Earphones?

- ☐ Yes
- ☐ No

If Answer is YES (Headphone Users)

5. Have you ever used headphones in the past?

- ☐ Yes
- ☐ No

6. For how long have you been using headphones?

- ☐ Less than 6 months
- ☐ 6 months – 1 year
- ☐ 1 – 3 years
- ☐ More than 3 years

7. On average, how many hours per day do you use headphones?

- ☐ Less than 1 hour
- ☐ 1 – 2 hours

- ☐ 2 – 4 hours
- ☐ More than 4 hours

8. Main purpose of using headphones? (Multiple responses allowed)

- ☐ Music
- ☐ Watching videos
- ☐ Online study / Work
- ☐ Phone calls
- ☐ Gaming
- ☐ While traveling

9. Type of headphones used

- ☐ Bluetooth Earbuds
- ☐ Wired
- ☐ Wireless

10. Do you usually listen at high volume?

- ☐ Never
- ☐ Rarely
- ☐ Sometimes
- ☐ Always

11. Do you receive mobile warning messages about high volume usage?

- ☐ Never
- ☐ Rarely
- ☐ Yes, sometimes
- ☐ Yes, frequently

12. What action do you take after receiving a warning message?

- ☐ Ignore the warning
- ☐ Reduce volume
- ☐ Stop using headphones for some time
- ☐ I have never received such warning

13. Have you experienced any problems due to headphone use? (Multiple responses allowed)

- ☐ Ear Pain
- ☐ Hearing Problem
- ☐ Headache
- ☐ Sleep Disturbance
- ☐ Stress / Irritation
- ☐ No problem

If Answer is NO (Non-Users)

14. Have you ever used headphones in the past?

☐ Yes

☐ No

15. If No, what is the reason for not using headphones?

☐ Health concerns

☐ Safety concerns during travel

☐ Not interested

☐ Cannot afford

☐ Other

16. Do you think headphone use affects travel safety?

☐ Yes

☐ No

☐ Not sure

17. Are you aware of health problems caused by excessive headphone use?

☐ Yes

☐ No